OmniWeave

Every frame, composed.

What if AI could direct a cinematic story — not just write text, but weave voice, images, music, and memory into a living experience?

Built for the Gemini Live Agent Challenge — Creative Storyteller

#GeminiLiveAgentChallenge

The Flat Chatbot vs. The Creative Director

	Standard AI	OmniWeave
Input	Single text prompt	Bidirectional 16kHz WebSocket audio + text prompt fallback
Output	Wall of text	Interleaved 24kHz multi-voice audio, 1K illustrations, cinematic video, and mood-synced ambient score
Continuity	Session Amnesia	Graph RAG Persistent Memory Bank Characters remember across stories
Execution	Single LLM	9 Gemini Models orchestrated via Google ADK

AI storytelling today is flat. Text in, text out. OmniWeave changes that.

Compose Mode: The Director's Pipeline

"A samurai cat defending a moonlit temple from shadow demons"

→

SequentialAgent

1. StoryWriter
`gemini-2.5-flash`
Drafts cinematic scripts with explicit image markers

↓

2. StoryReviewer
`gemini-2.5-flash-lite`
Validates narrative consistency and speaker label formatting

→

Visuals
`gemini-3.1-flash-image-preview`
1K 16:9 illustrations or veo-3.1 video clips

Story DNA
`gemini-embedding-2-preview`
Multimodal vector extraction for similarity

Audio
`Cloud TTS Chirp 3 HD`
28 Multi-voice acting, gender/age-matched

→

Cinematic Experience

~30 seconds

Engineering Insight — Why ADK SequentialAgent? The reviewer agent guarantees quality control — catching inconsistencies and validating character sheet adherence before any visual or audio models are triggered.

Anatomy of a Scene

1

Visual Grounding

`gemini-3.1-flash-image-preview`

Generates 1K, 16:9 images with cinematic Ken Burns auto-panning. Every prompt is completely self-contained to prevent visual drift.

3

Ambient Score

`lyria-realtime-exp`

Generates mood-reactive background music mapped from story keywords (e.g., "epic orchestral" for battle scenes).

2

Multi-Voice Casting

`Cloud TTS Chirp 3 HD`

Intelligently maps 28 HD voices by gender/age heuristics. Kenji sounds completely different from Kiko.

4

Caption Control

Hide captions to reveal edge-to-edge art, show them to read along like movie subtitles. Color-coded speaker badges.

Live Mode: Real-Time Bidirectional Orchestration

1. User speaks (16kHz PCM audio captured by browser)
2. Cloud Run WebSocket Proxy streams to gemini-2.5-flash-native-audio
3. AI responds instantly with expressive 24kHz voice narration + real-time text
4. Client-side triggers image + music generation in background
   Non-blocking — audio flow is never interrupted

ON AIR Interruptible
Natural conversation flow — interrupt the AI mid-sentence and it adapts.

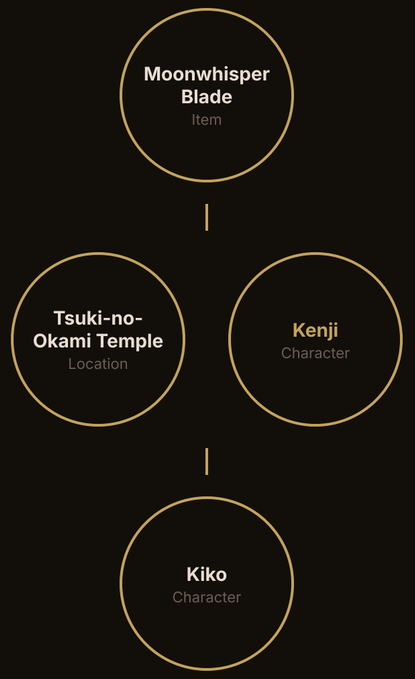
Unlimited Sessions
Sliding context window compression (100K trigger, 80K target) prevents session crashes.

Auto-Reconnect
Session resumption + WebSocket keep-alive ping every 15s for stability.

Audio Transcription
Input + output audio transcribed for the visual transcript log.

Director's Note: Native audio models drop context when forced to call tools mid-conversation. OmniWeave compensates via client-side trigger orchestration, ensuring continuous audio flow while illustrations render in the background.

The Memory Bank: Persistent World-Building



Automatic Entity Extraction

`gemini-2.5-flash`

Extracts characters, locations, and relationships via JSON schema-enforced structured output.

Persistent Knowledge Graph

Builds a Firestore-backed graph of nodes (characters, locations, items) connected by relationship edges (allies, visits, wields).

Cross-Story Continuity

OmniWeave remembers exact appearances and traits from the Memory Bank. A samurai cat from last week's story shows up with the same emerald eyes and crimson obi.

Zero-Friction Auth

Firebase anonymous auth + Google Sign-In. Multimodal embeddings compute story fingerprints for similarity discovery.

The Orchestration: 9 Models. 1 Vision.

Why 9 models? Because no single model is best at everything. **OmniWeave acts as the Director**, casting the perfect model for every specific task.

`gemini-2.5-flash`
StoryWriter + Entity Extractor
~6s story generation

`gemini-2.5-flash-lite`
StoryReviewer
~1s quality gate

`gemini-3-flash-preview`
Director Agent
ADK root agent orchestration

`gemini-2.5-flash-native-audio`
Live Bidi-Streaming
Real-time voice narration

`gemini-3.1-flash-image`
1K Illustrations
~12s per scene

`veo-3.1-generate`
Cinematic Video Clips
~27s per clip with audio

`gemini-2.5-pro-tts`
Fallback Voice Acting
Multi-speaker TTS

`gemini-embedding-2`
Story Fingerprints
3072-dim multimodal vectors

`lyria-realtime-exp`
Mood-Aware Ambient Score
Real-time music generation

Built & Shipped on Google Cloud

Compute	Data & Media	Delivery & Auth	Intelligence
Cloud Run ADK agent server + WebSocket proxy	**Cloud Firestore** Story storage, users, Graph RAG	**Firebase Hosting** React 19 SPA frontend	**Cloud Text-to-Speech** Chirp 3 HD engine (28 voices)
Cloud Build CI/CD pipeline via cloudbuild.yaml	**Firebase Storage** Full-res media assets (images, audio, video)	**Firebase Auth** Anonymous + Google Sign-In	**Artifact Registry** Docker image repository

LIVE PRODUCTION READY `deploy-all.sh executing... cloudbuild.yaml triggered... omniweave-adk deployed successfully`

OmniWeave proves that AI can be more than a text box.

It can be a creative director, a narrator, a composer, and a visual artist — all at once.

Live: gen-lang-client-0001923421.web.app

Code: github.com/musicksto/Omni_weave

ADK Server: omniweave-adk-54597885936.us-central1.run.app

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